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Structure of stagnated plasma in aluminum wire array Z pinches

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Experiments with aluminum wire array Z pinches have been carried out on the mega-ampere generator for plasma implosion experiments (MAGPIE) at Imperial College London [I. H. Mitchell *et al.*, Rev. Sci. Instrum. **67**, 1533 (1996)]. It has been shown that in these arrays, there are two intense sources of radiation during stagnation; Al XII line emission from a precursor-sized object, and both continuum and Al XIII radiation from bright spots of either significantly higher temperature or density randomly distributed around this object so as to produce a hollow emission profile. Spatially resolved spectra produced by spherically bent crystals were recorded, both time-integrated and time-resolved, and were used to show that these two sources of radiation peak at the same time. © *2006 American Institute of Physics*. [DOI: 10.1063/1.2234284]

I. INTRODUCTION

The wire-array Z pinch is the world's most powerful x-ray source, and thus of interest to the Inertial Confinement Fusion (ICF) community as a radiation source for indirect drive ICF concepts.[1–3] Soft-x-ray yields of up to 1.8 MJ have been achieved, and powers of 280 TW have been produced with nested wire arrays on the Z accelerator at Sandia National Laboratories:[4] a 20 MA pulsed-power current generator with a rise time of 100 ns.[5] However, the mechanism by which these short, intense bursts of radiation are produced is still open to question. With the next generation of wire-array Z-pinch generators on the horizon, the physics of x-ray production demands attention, offering the possibility of increasing x-ray power and yield as well as the tailoring of pulse shape, which is of importance for ICF.

Despite intense study of the possible contributions of kinetic energy, compressional heating, and ohmic heating to radiation production, a satisfactory explanation of the precise timing, structure, and power of the x-ray pulse remains elusive. Experiments[6–8] have shown that wire array implosions proceed as a complex process of steady ablation from the wires, filling the array with plasma, leading to wire breakage[9] and, finally, one or more snowplough implosions,[10] so named due to the manner in which the imploding current-carrying sheath accumulates plasma already present inside the array as it collapses. Additional complexity arises with the observation that the snowplough implosion does not involve 100% of the array mass; up to 40% of the array material remains near the original wire positions while the first implosion proceeds.[9,11,12] Current, carried towards the axis during an implosion, may switch back out to flow through this "trailing mass" at some point during the stagnation phase. Such a "current restrike"[10] can produce another implosion, resulting in the trailing mass being swept inwards onto the stagnated plasma of the first implosion.

The complexity of these implosion processes, particularly their vulnerability to a number of instabilities [such as a Rayleigh-Taylor-like instability in the imploding sheath, and magnetohydrodynamic (MHD) instabilities in the stagnating plasma] make the understanding of x-ray production a daunting task. Instruments that are able to provide information on the spectral and spatial structure of plasma emitting in the 1–20 Å (600 eV–12.4 keV) range are of great benefit to the diagnosis of such high-energy high-density plasmas. Diagnostics utilizing spherically bent crystals to provide both spectrally and spatially resolved imaging systems were first developed in the 1970s.[13,14] They have since been used in the investigation of laser-produced plasmas,[14] exploding-wire Z pinches, and X pinches,[15,16] and are now becoming an exceptionally useful variety of instrument in the study of wire-array plasmas.[17,18] The focusing nature of these devices results in a much higher output intensity than can be accomplished with flat or cylindrically bent crystal spectrometers, often rendering spherically bent crystal instruments the only option in circumstances in which the source intensity is too weak for other diagnostics to produce a workable output. In a situation such as discussed in this paper, where the source intensity is relatively strong, the high sensitivities achievable with spherically bent crystals allows these instruments to be placed far from the source. This is of great advantage, since wire arrays produce a large amount of high-velocity debris, capable of damaging delicate instruments be-

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yond repair. Having the ability to place these instruments far enough from the source such that debris shields can provide sufficient protection against damage, while still producing a strong signal, makes fielding these diagnostics viable.

Combined with the ability to achieve spectral resolutions up to $\lambda/\delta\lambda=10000$ and spatial resolutions down to a few micrometers, spherically bent crystal spectroscopy may prove indispensable to understanding x-ray production in wire arrays.

In this paper, experiments on the MAGPIE generator (1 MA, 240 ns rise time) at Imperial College London[19] are described, in particular the structure of stagnated plasma and x-ray production in aluminum wire arrays. In addition to the suite of diagnostics used in previous experiments on MAGPIE, described in Sec. II, a new spectrometer was fielded that is capable of utilizing four (or more) spherically bent crystals simultaneously. For each experiment, two mica crystals and one quartz crystal were employed to produce time-integrated axially resolved spectra, time-resolved measurements of x-ray intensity over ten spectral windows for an axial section of the pinch, and time-integrated axially and radially resolved monochromatic images. Large, unknown rates of self-absorption and reemission of aluminum line radiation makes quantitative spectroscopy highly problematic. However, the spatial resolution achieved by this instrument, when combined with data from other diagnostics, has provided new insights into the spatial and temporal origin of both continuum and line x-ray emission.

In Sec. II, the diagnostic suite is described in detail, including a description of both the concept of spherically bent crystal spectrometry as well as the diagnostic itself. The results from the first experimental series utilizing this new instrument are presented and discussed in Sec. III. Finally, a summary of the conclusions that might be drawn from these experiments is presented in Sec. IV, including possible directions for future work.

II. EXPERIMENTAL SETUP

The results discussed in this paper all relate to $N=32$ wire arrays of 10 μm aluminum, 23 mm in length, with a diameter of 16 mm. The total mass of each array was 67.9 μg/cm. These particular specifications were chosen because previous experiments on the MAGPIE generator have shown the x-ray pulse from these arrays to be highly reproducible.

Figure 1 shows the layout of the various diagnostics, described below, which were fielded during these experiments. Several different pinhole imaging systems, both time-resolved and time-integrated, were implemented on these experiments: Two side-on (r, z) XUV framing cameras (four individually gated MCP sections, each imaged onto by a 100 μm pinhole) provided time-resolved images (3 ns gating time) in the XUV region when operated without any filtering, and for radiation >500 eV when filtered with 7 μm beryllium.

Two side-on time-integrated pinhole cameras, labeled A and B (diagnostics 7 and 9, respectively, in Fig. 1), were fielded to provide imaging at shorter wavelengths. Camera A, operating at a magnification of 0.54, was equipped with seven 20 μm pinholes, filtered with 12.5 μm titanium, 20 μm beryllium, 15 μm silicon, 12.5 μm PVDC+1 μm aluminum, 6 μm aluminum, 6 μm iron, and 250 μm beryllium. Camera B, operating at a higher magnification of 1.26, utilizsed three 50 μm pinholes, filtered with 6 μm aluminum, 250 μm beryllium and 15 μm silicon. The choice of these filters was based on the same principle as a Ross pair filter set, in which neighboring elements in the same period are used as filters—the similarity between the transmissions allowing subtraction of one filtered signal from the other to provide a very narrow band pass. In the case of the common filters on cameras A and B (aluminum, silicon, and beryllium), the subtraction of the aluminum and beryllium signal from the silicon results in a band pass window between $\approx$1.56 and 1.84 keV, which provides an isolated signal of the aluminum XII lines at $\approx$1.6 keV. Cameras A and B were arranged 90° apart azimuthally to allow analysis of pinch structure as viewed from two different angles, and produce pairs of images that, through parallax, provide information as to whether certain features are located on or off axis.

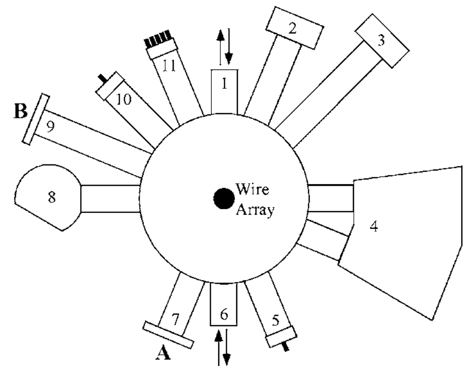

FIG. 1. The layout of diagnostics used in the experiments: (1) and (6) double-pass laser probing, (2) and (3) XUV framing cameras, (4) spherically bent crystal spectrometer, (5) single silicon diode, (7) time-integrated pinhole camera A, (8) convex mica crystal x-ray spectrometer, (9) time-integrated pinhole camera B, (10) single PCD, and (11) 5 PCD pack.

Six filtered diamond PCDs (photoconducting detectors) were fielded to provide time-resolved x-ray power in a number of wavelength bands of interest, 1.5 and 6 μm polycarbonate filters were used to observe soft-x-ray power, while 6 μm aluminum, 12.5 μm PVDC, 12.5 μm titanium, and 6 μm iron filters were employed to monitor progressively harder x-rays. A single silicon diode, filtered with 180 μm aluminum, was employed to monitor x-ray power above 6 keV.

A convex mica crystal x-ray spectrometer provided either time (and spatially) integrated spectra when backed with film, or time-resolved (and spatially integrated) x-ray power over a narrow wavelength range when backed with a single photoconducting diode.

An ND-YAG laser, frequency doubled to 532 nm and pulse compressed to 0.4 ns, provided schlieren imaging, shadowgraphy, and interferometry of the implosion where the plasma density was low enough such that $n_e < 10^{19}$ cm^{-3}. A double-pass laser system was employed: a single laser pulse was split into two beams, one of which

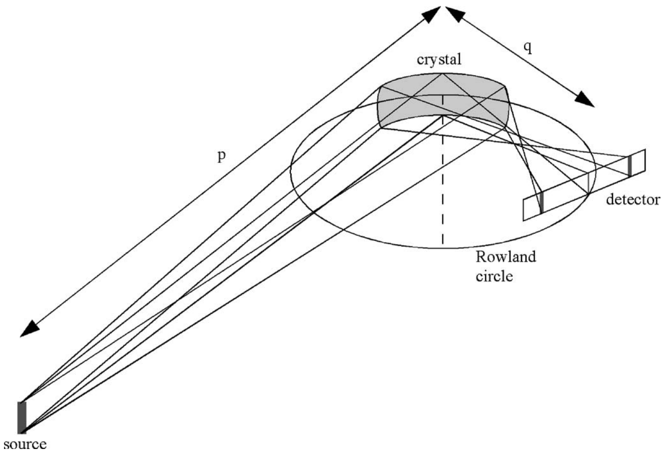

FIG. 2. The spatial focusing of a spherically bent crystal in the plane perpendicular to the Rowland circle (sagittal plane).

passed immediately through the load chamber in one direction, while the other followed a longer path (creating a delay of 23 ns) and passed through the load chamber in the opposite direction. An optical streak camera, sampling slit aligned radially, recorded the trajectory of the implosion. A more detailed description of the instrumentation used in these experiments can be found in Refs. 20 and 21 and references therein.

A spherically bent crystal spectrometer, operating in a number of varying modes, provided spectral information with spatial resolution, and is described in detail below.

A. Spherically bent crystal spectrometry

The development and operation of these instruments, often referred to as FSSRs (focusing spectrographs with spatial resolution), is well described in Refs.14, 17, and 18. The manner in which this type of instrument produces both spectral dispersion and spatial focusing may be understood through Figs. 2–5, which illustrate the different aspects of a typical spherically bent crystal imaging system.

A source, in this case a plasma, emits radiation that is incident upon the surface of a spherically bent crystal, with a curvature radius R. A useful aid in the description of spherically bent optics is the Rowland circle, a circle with radius $R/2$ that intercepts the center point of the crystal and whose diameter is aligned tangentially to the surface of the crystal at this center point. Figure 2 shows how the crystal forms an image of an x-ray source in the sagittal plane, the plane perpendicular to the Rowland circle (described in detail in Ref. 14). Radiation undergoes Bragg reflection (and thus dispersion) from the crystal in the meridional plane of the spherical surface (the same plane as the Rowland circle), and at the same time is spatially focused in this plane. Whereas each point of the source radiates a range of wavelengths, all of which are incident over the entire surface of the crystal, the only wavelengths that are reflected are those that satisfy the Bragg condition at each point on the crystal,

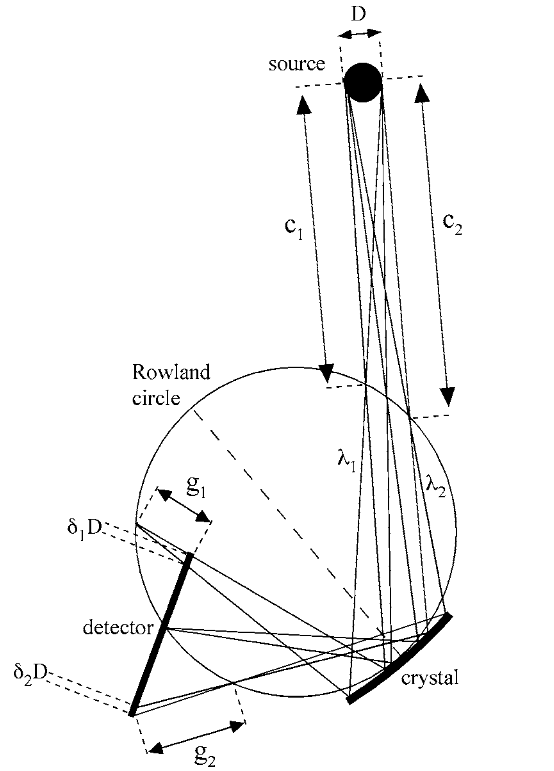

FIG. 3. When the source has finite size and the detector does not lie directly on the Rowland circle, the image has spatial resolution in the meridional plane.

$$2d \sin \theta = m\lambda, \tag{1}$$

where the angle θ is the grazing angle between the incident ray and the crystal surface at the point of incidence (illustrated in Fig. 5), d is the interplanar spacing of the crystal, and m is an integer corresponding to the order of reflection. Through Eq. (1), each point of the crystal in the meridional

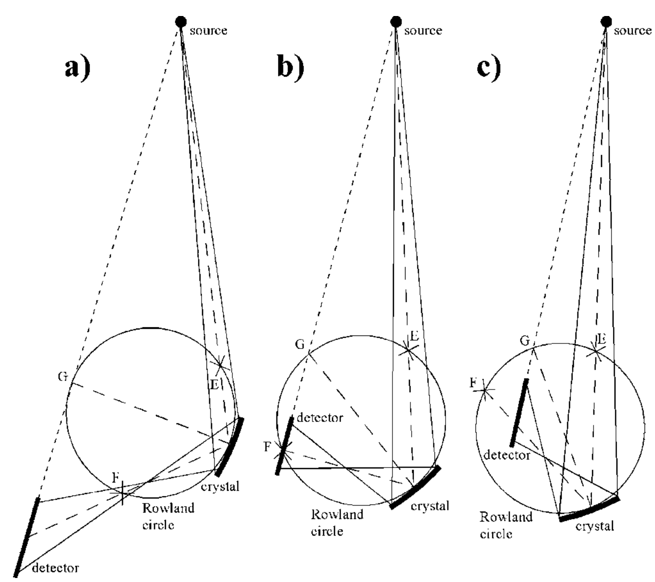

FIG. 4. Three possible scenarios for crystal/detector setup that illustrate how the detector can be (a) completely outside the Rowland circle, (b) cross the Rowland circle at some point, or (c) be completely inside the Rowland circle.

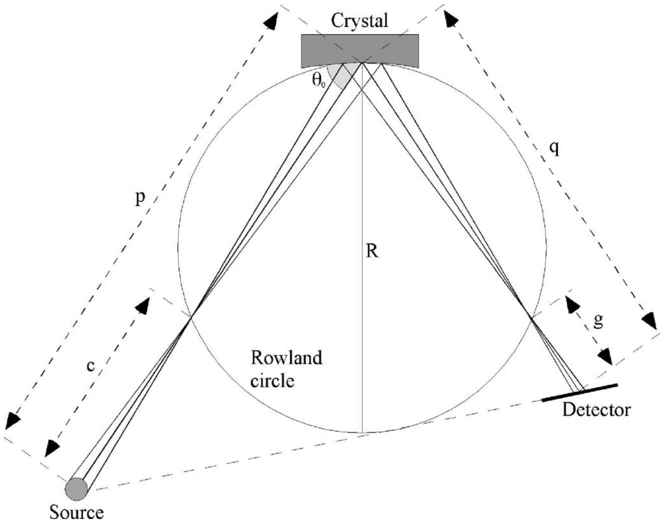

FIG. 5. An example of magnification in the meridional plane for one particular wavelength.

plane corresponds to the reflection of a unique wavelength for each order of reflection.

Spatial focusing by a spherical surface in the sagittal plane is achieved according to the lens formula,

$$\frac{1}{p}+\frac{1}{q}=\frac{1}{f_s}, \tag{2}$$

where p and q are the distances from the source to the crystal and the detector to the crystal, respectively, as shown in Fig. 2, and f_s is the focal length in the sagittal plane,

$$f_s=\frac{R}{2\sin\theta}. \tag{3}$$

If a detector is placed such that it crosses the Rowland circle, then it will not suffer any reduction in spectral resolution due to image extent in the meridional plane, but only at the point at which it crosses the Rowland circle. At this particular point, the image will only have spatial resolution in the sagittal plane, whereas all other points on the detector suffer spatial broadening in the meridional plane, which limits the spectral resolution. This effect is shown in Fig. 3 and is a result of the source having finite size D in the meridional plane. Figure 3 illustrates how two regions on the detector (one inside the Rowland circle and one outside) suffer spatial broadening such that wavelengths λ_1 and λ_2 are spread across distances $\delta_1 D$ and $\delta_2 D$. For a given wavelength, the ratio of the distance the detector is away from the Rowland circle (g_1 and g_2 in Fig. 3) to the distance the source is away from the Rowland circle (c_1 and c_2 in Fig. 3) gives the magnification of the image in that wavelength. This relation is shown explicitly later, in Eq. (6), where its relevance to 2D imaging is discussed.

If spectral resolution is important for some specific wavelength, the distance q should be calculated such that the detector crosses the Rowland circle at this wavelength. This fixes q and therefore means that the focusing condition of Eq. (2) is only possible by varying the distance from the source to the instrument such that

$$p=\frac{R\sin\theta}{2\sin^2\theta-1}. \tag{4}$$

Of course, this condition can only be met at one particular meridional point on the detector, where it crosses the Rowland circle. All other points will have defocusing δD in the spectral direction, the severity of which depends on how far from the Rowland circle these points lie.

The freedom to vary the distance from the source to the crystal is often not achievable in experiments in which this distance is either fixed or has a limited range due to physical constraints on the equipment, and instead the crystal-detector distance is the variable used to achieve focusing. Both the distance from the source to the crystal and the Bragg angle determine whether the detector will need to be placed inside, on, or outside the Rowland circle to achieve focusing. The three possible locations for a detector (inside, outside, or crossing the Rowland circle) are shown in Fig. 4. For $\theta<45°$, the detector will always lie outside the Rowland circle, no matter the source distance, and this is illustrated in Fig. 4(a). For this situation, there will be significant spatial broadening across the entire spectrum. When $\theta>45°$, then for a critical combination of θ and p the detector is positioned such that it crosses the Rowland circle [shown in Fig. 4(b)] and the zero spatial broadening condition described by Eq. (4) can be met for one wavelength. As θ or p increase outside of the critical combination (i.e., as the source moves further away or the Bragg angle increases so as to become closer to being normal), the detector moves inside the Rowland circle, and this is illustrated in Fig. 4(c). Such an arrangement is not good for performing spectroscopy, since the spectral band reflected by crystals at large Bragg angles is very narrow. It is good, however, for performing monochromatic imaging, and the reasons for this will be discussed shortly.

The property of the crystal to image points on the incident radiation side of the Rowland circle onto corresponding points on the opposite side of the Rowland circle is shown in Fig. 4: Point E, which lies on the intersection of the Rowland circle and the line between the source and the center of the crystal, is imaged onto point F. The position of point F is the reflection of point E in the line which bisects the Rowland circle, connecting the center of the crystal and point G.

As discussed, points on the detector that do not lie directly on the Rowland circle have spatial resolution in both the sagittal and meridional directions. For analysis of spectral features this is most often a disadvantage, as the spatial broadening reduces the spectral resolution of the system. However, when a full 2D image of the plasma is sought, the detector is deliberately placed such that it does not cross the Rowland circle. To achieve full focusing in both the sagittal and meridional directions requires that θ be very close to 90°, since the spatial focusing characteristics of a spherically bent crystal are the same as those of a spherically concave mirror: The magnification in the sagittal direction is given by

$$\sigma=\frac{q}{p}=\frac{R}{2p\sin\theta-R} \tag{5}$$

while magnification in the meridional plane is given by

$$\nu = \frac{g}{c} = \left| \frac{R[p - \sin\theta(2p\sin\theta - R)]}{(R\sin\theta - p)(2p\sin\theta - R)} \right|. \quad (6)$$

An example of magnification in the meridional plane as a result of $g \neq 0$ is shown in Fig. 5 for one particular wavelength. Comparison of Eqs. (5) and (6) show that unless $\theta = 90°$, the magnifications in the two planes are different, which is a result of the crystal having different focal lengths in the sagittal and meridional planes when $\theta \neq 90°$. This means that when $\theta \neq 90°$, spatial focusing can only be achieved in one direction, with the other direction defocused. When $\theta = 90°$, however, the sagittal and meridional focal lengths (and thus magnifications) become equal and spatial focusing can be achieved in both directions simultaneously.

At a point L on the detector upon which a ray of wavelength λ is incident, the linear dispersion (i.e., per unit length "l" along the detector in the meridional plane) is given by

$$\frac{d\lambda}{dl}(L) \propto \cot\theta. \quad (7)$$

From Eq. (7) it can be seen that dispersion tends to infinity as $\theta \to 90°$, meaning that the imaging system effectively becomes monochromatic when $\theta = 90°$, for which $\lambda = 2d/m$.

To summarize, when θ is close to 90°, a single, fully focused 2D image in near-monochromatic radiation is achievable (experiments by Pikuz *et al.* have shown that well-focused 2D images can be produced with θ between 82° and 88°[22]). For achieving a spectrum, however, θ is reduced and the detector is placed to achieve focusing in the sagittal plane in accordance with Eqs. (2) and (3), and the finite dispersion results in a spectrum with fully focused spatial resolution in the sagittal direction and a variable degree of defocused spatial resolution in the meridional direction. The dependence of f_s with θ demands that in order for focusing in the sagittal direction to be maintained across the extent of the detector, the detector must be aligned as shown in Fig. 4: The surface of the detector must lie along the line in the meridional plane that connects the source and point G (i.e., once the crystal to detector position has been calculated, the detector must be angled such that its surface creates a line that intercepts the source).

B. Implementation of a spherically bent crystal spectrometer on the MAGPIE generator

An instrument capable of operating multiple crystals simultaneously in the modes described in the previous section was designed and built for use on the MAGPIE generator and is illustrated in Fig. 6. The vacuum chamber that houses the crystal diagnostics occupies one octant of the instrument platform surrounding the load chamber, and has two direct horizontal lines of sight to the pinch, which provide all the crystals with an unobstructed view of the entire length of the pinch.

Two mica crystals ($2d = 19.88$ Å), with curvature radii $R = 186$ and 100 mm, were operated in the $\theta < 90°$ regime; producing spectra with sagittal focusing along the axis of the load. Mica has very high reflectivity in ten orders

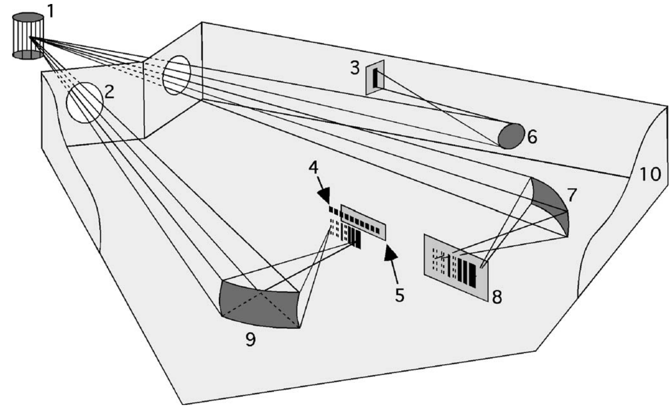

FIG. 6. Spherically bent crystal spectrometer: (1) wire array, (2) apertures—covered with 3 μm aluminized mylar plus 16 μm polypropylene as debris protection, (3) monochromatic imaging film pack, (4) silicon detector array, (5) film pack—backing detector array, (6) quartz crystal, $R = 250$ mm, (7) mica crystal, $R = 186$ mm, (8) film pack, (9) mica crystal, $R = 100$ mm, (10) vacuum chamber.

($m = 1-5, 7, 8, 11-13$) which allows spherically bent mica crystals to operate effectively between 1.1 and 19.88 Å. Both mica crystals were aligned such that θ_0 (the Bragg angle at the center of the crystal) was 54°, producing a second-order ($m = 2$) reflection centered on ~8 Å and imaging the K-shell lines of Al XII.

The detector for the $R = 186$ mm crystal was a film pack loaded with Kodak DEF film and filtered from UV and visible radiation with 40 μm beryllium, producing time-integrated spectra. The positioning of the crystal and film pack produced a spectrum from ~7.4 to 8.5 Å in second-order reflection, and ~4.5 to 5.7 Å in third-order reflection. The source to crystal distance was maintained at 815 mm and so, from Eq. (4), the film pack crossed the Rowland circle at the point at which radiation of wavelength 7.6 Å was reflected in second order. Thus, the section of the spectrum that corresponds to the second-order wavelength of 7.6 Å suffered no spatial broadening.

The primary detector for the $R = 100$ mm crystal was an International Radiation Detectors AXUV-10EL silicon photodiode array.[23] The placement of this detector array is illus-

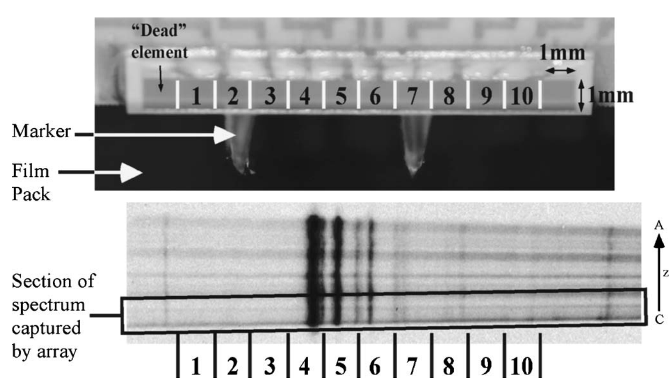

FIG. 7. The silicon x-ray detector array. An $R = 100$ mm mica crystal images an axial section of an FSSR 1D spectrum onto the array, and markers beneath cast a shadow onto the film pack behind, enabling the array elements to be aligned with the full spectrum imaged by the $R = 186$ mm crystal.

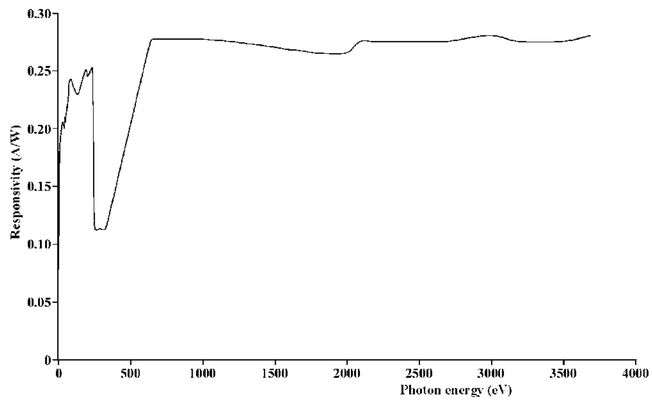

FIG. 8. The response of the photodiode array.

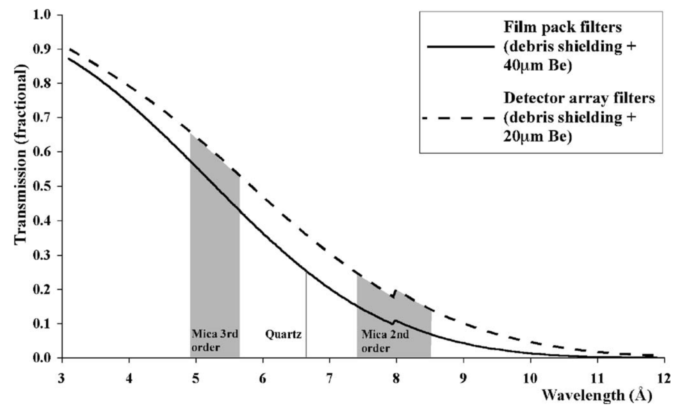

FIG. 9. The variation of transmission fraction with wavelength for the crystal filters, which total 2 μm of aluminized mylar and 16 μm of polypropylene as debris shielding. Filtering from uv-visible for the detector array and the film packs adds an additional 20 and 40 μm of beryllium, respectively.

trated in Fig. 6. The detector itself is shown in Fig. 7: the small squares labeled 1 to 10 in the center of the upper photograph are the elements of the detector, each one an x-ray detector. The PCB upon which the detector was mounted can be seen in the background at the top of Fig. 7, while at the bottom of this photograph a pair of plastic markers can be seen hanging immediately beneath the detector elements. The detector array consists of ten 1×1 mm silicon x-ray photodiodes, with a rise time of 1 ns and a flat response curve from $\sim$700 to 4000 eV, as can be seen from Fig. 8 (taken from Ref. 23). The voltage signals produced by the elements of the silicon array are comparable to the signals produced by the PCDs. By aligning the elements of the array in the meridional plane, each element provided a measurement of time-resolved x-ray power over a particular spectral range. To ascertain the spectral range covered by each element, a film pack (loaded with Kodak DEF film) was placed immediately behind the detector array and a pair of markers was fixed beneath the array. The height of the spectrum in the sagittal direction allows approximately one-third of the image to be captured by the array (shown in Fig. 7) while the remaining two-thirds is recorded by the film pack, including the shadows cast by the markers. The position of the marker shadows on the film image can then be used to calculate the spectral range covered by each element, and these data compared with the full spectrum imaged by the $R=186$ mm crystal. The detector array was positioned to capture a section of the spectrum between $\sim$7.5 and 8.2 Å in second-order reflection. The position of the detector array was changed from shot to shot in order to capture individual spectral lines on separate elements, as shown in Fig. 7. After alignment, both the array and film pack were sealed inside a lightproof box (with an aperture covered with 20 μm beryllium) as the detector array has some optical sensitivity.

A quartz crystal ($2d=6.66$ Å) with curvature radius $R=250$ mm was operated in the $\theta\sim90°$ regime, producing fully resolved 2D monochromatic images at a central angle $\theta_0\sim88°$, giving $\lambda\sim6.66$ Å. For crystals such as mica, which Bragg-reflect in many different orders,[24] it is impossible to identify which orders are responsible for an image produced in the $\theta\sim90°$ regime. Quartz exhibits (practically) first-order reflections only. The detector for the quartz crystal was a film pack, loaded with Kodak DEF film and filtered from UV and visible radiation with 40 μm beryllium.

Alignment is achieved with a laser that is placed, by way of an optic fiber, in the center of the load chamber prior to the insertion of the wire array. Although alignment is checked before each shot, neither the vacuum chamber in which the crystals are mounted, nor the crystals themselves, are moved between experiments, which allows the angle of the crystals to remain identical from one experiment to another. In order to protect the crystals from debris produced during experiments, the apertures through which the vacuum chamber is connected to the load chamber are sealed with several layers of debris shielding, which amount to 2 μm of aluminized mylar and 16 μm of polypropylene. The debris shielding absorbs a certain amount of radiation in the wavelength range imaged by both the mica and quartz crystals. In addition to this, there is absorption from the beryllium filters covering the film packs and the detector array. For the film pack that images the 186 mm mica crystal spectrum, the total transmission fraction of the filter set for second- and third-order reflections across the spectrum ranges from $\sim$0.07 to 0.15 and $\sim$0.43 to 0.57, respectively. The quartz crystal film pack has an identical set of filters, and the transmission fraction is $\sim$0.25 for monochromatic imaging. The thinner beryllium used to lightproof the detector array results in a slightly higher transmission fraction; $\sim$0.14 to 0.24 and $\sim$0.53 to 0.66 for second- and third-order reflections, respectively. The variation of transmission fraction with wavelength as a result of these filter combinations is illustrated in Fig. 9. It should be noted that the transmission fractions discussed here are a result of the filters only and are mentioned purely to illustrate the typical losses resulting from debris shielding and the need to filter from UV and visible radiation. The total transmission fraction from source to detector must include the reflectivity of the crystal, which is a function of curvature radius, Bragg angle, incident wavelength, and any aberrations of the crystal surface. Hence, the reflectivity is highly specific to each crystal and must be individually measured across all wavelengths and angles. This mea-

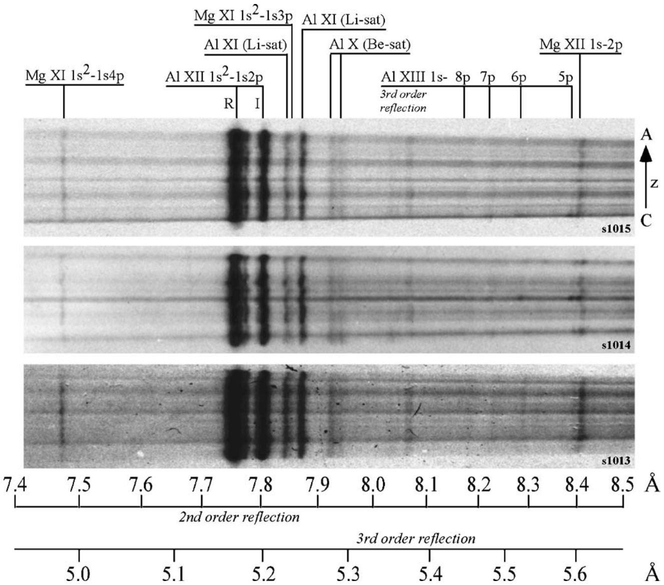

FIG. 10. Typical crystal spectra from 32 wire, 10 μm aluminum arrays. All spectral lines are a result of second-order Bragg reflection, unless labeled otherwise.

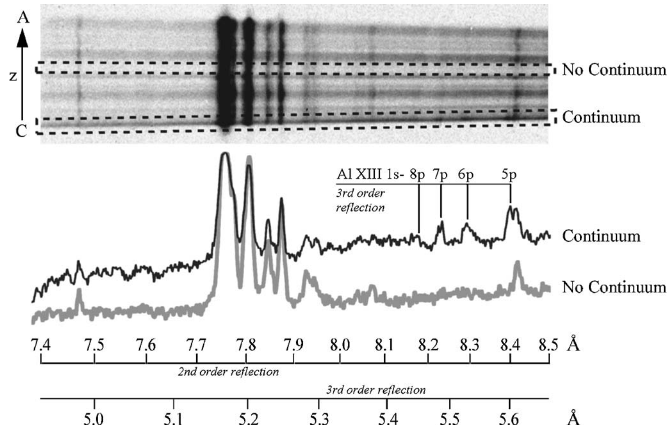

FIG. 11. Lineouts showing the variation of intensity with wavelength for two axial regions, one with continuum radiation and the other without any continuum radiation.

surement is difficult to perform, and consequently the reflectivities of the crystals used in these experiments are unknown.

III. DISCUSSION OF RESULTS

A. Time-integrated spectra

Figure 10 shows several typical time-integrated spectra produced by the 186 mm mica crystal, imaged onto Kodak DEF film. In these images, the vertical direction is parallel to the pinch axis and spatially resolved, while wavelength is dispersed along the horizontal direction. The orientation of the anode (labeled "A") and cathode (labeled "C") is also shown. Although the dominant feature on all these spectra is the He-like aluminum resonance line (Al XII *K*-shell) at 7.756 Å, lines from several other ion species are clearly visible, including Mg XII and Mg XI (lines at 7.5 and 8.42 Å). The position of these lines relative to one another allows the wavelength scale of each spectrum, included for second- and third-order Bragg reflections on Fig. 10, to be calculated. All the lines identified in Fig. 10 are second-order Bragg reflections unless specifically labeled otherwise.

A striking feature of these spectra is the presence of continuum radiation from narrow sections of the pinch, appearing as horizontal features on the spectra. Two mechanisms contribute to the production of continuum radiation: bremsstrahlung (free-free) radiation and recombination (free-bound) radiation. The relative contributions of these two mechanisms to the total continuum spectrum are a function of both temperature and density for a particular plasma.

The fact that the features in Fig. 10 are continuum radiation, as opposed to line radiation, makes it impossible to characterize the intensity versus wavelength profile from this type of spectrum: any point on one of these continuum features could be the result of many different orders of Bragg reflection, but there are no features to distinguish which orders are present or their relative strength. A typical shot produces about three to five narrow regions (randomly spaced along the axis, with Δz between 0.4 and 1.6 mm, with a mean of $\Delta z \approx 0.9$ mm) which emit intense continuum radiation, and approximately another three to four slightly wider regions that also emit in the same manner, although less intensely.

At points on the spectra where continuum radiation regions cross line radiation, the line radiation is much more intense. This is not simply a result of unrelated uniform line radiation and continuum radiation adding together to darken the film at locations where both are present. Adjusting the gamma levels of the scanned film reveals that film darkening at the crossover points is much more than the darkening that results from overlapping one region with continuum and no line radiation and another region with line radiation and no continuum (assuming the response of the film is linear over these intensities). For some shots, the intensity of emission was sufficient to overexpose the film at axial positions where the Al XII *K*-shell line corresponded with continuum radiation. Despite this, it is still very clear from the spectra in Fig. 10 that radiation from this line (and most of the other lines visible in these spectra) is emitted from the entire length of the pinch, as line radiation is present at axial positions where there is no continuum.

However, there are some lines whose radiation can only be seen at axial positions where continuum is present. This can be seen in Fig. 11, where a lineout has been taken along the length of the spectrum at two axial positions: one where there is intense continuum radiation, and another where there is no continuum. The lines that are only observed in the continuum regions are imaged between 5.4 and 5.6 Å as a result of third-order Bragg reflection and originate from hydrogen-like aluminum (Al XIII), while lines present in both regions only originate from less ionized species. The presence of Al XIII in one region of the stagnated column (the "bright spots") but not another region could be a result of either higher temperature or higher density in the bright spots than elsewhere.

The energy required to ionize Al XII to Al XIII is

$\approx$2.1 keV (a significantly larger energy than the $\approx$440 eV required to ionize Al XI to Al XII). Thus, if high temperature were solely responsible for the increased ionization, the electron temperature in a bright spot would have to be high enough for this $\approx$2.1 keV ionization energy to be achieved by a significant fraction of the ions in the bright spot. The electron temperature outside the bright spots need only be sufficient to ionize from Al XI to Al XII.

It is important to realize that the difference in thermal conditions that determine whether a species is actually produced is much greater than the difference determining whether there is sufficient excitation energy to produce the lines. In this case, comparing the bright spots to the bulk plasma, the ionization energy required to produce Al XIII ($\approx$2.1 keV) is approximately 1.66 keV higher than the ionization energy required to produce Al XII ($\approx$440 eV). However, for highly ionized species, even the lowest energy line transition (from $n=2$ to $n=1$) requires an excitation energy of approximately 3/4 the ionization energy of the ion. For aluminum, the lowest energy line transition for Al XIII is $\approx$1.73 keV, while for Al XII it is $\approx$1.59 keV. From this it can be seen that there is only 140 eV difference between the excitation energy needed to excite line transitions in Al XII and Al XIII, whereas the difference in ionization energy required to actually produce these ions is 1.66 keV, more than 10 times higher. Therefore, it is conditions that might produce different levels of ionization, not excitation, that must be considered when lines from a species are seen in one location but not another in a situation such as this.

Increased levels of ionization can also result from high density by means of a ladder ionization process, discussed in detail in Refs. 25 and 26 and references therein: as density increases, so too does the opacity of the plasma to line radiation, effectively resulting in photon pumping of the excited states of the various ion species. The ionization energy for an ion in an excited state is significantly lower than that of the ion in the ground state, meaning that ionization occurs more readily for ions in an excited state. Thus, with a greater population of excited states brought about by higher density (and thus opacity), the average ionization state of the plasma increases.

Both the free-free and free-bound mechanisms by which continuum radiation is formed have a strong density dependence (n^2) due to the fact that they are two-body interactions. Therefore, since intense continuum radiation is a signature of high density, and highly ionized ion species can result from high densities via the ladder ionization process, it might be the case that the bright spots observed in the mica spectra primarily occur due to higher density rather than higher temperature. This is not to say that the bright spots might not exhibit high temperatures, but that high density rather than high temperature might be the dominant factor with regard to the production of radiation from these regions.

To obtain further insight into the high-density versus high-temperature issue, the spectra produced by cylindrical Al plasmas of diameter 1 mm, similar to the stagnating columns observed in these experiments at the time of maximum compression and peak x-ray power, were calculated. These calculated spectra provide quantitative guidance regarding the changes in line and continuum intensities brought about by density and/or temperature changes in the plasma column. The calculations have been carried out using a detailed configuration atomic model coupled self-consistently to full radiation transport (see Sec. 4 of Ref. 27), with enough frequencies (2277) in the grid to adequately follow each line profile. Figure 12 shows the results of these calculations for a region of the spectrum captured in part by the mica crystals. A feature of particular interest is the Al XIII δ line at 2212 eV (5.6 Å), which is one of the lines that is only observable where there is intense continuum radiation in the mica spectra. The Al XII resonance and intercombination lines, the dominant features of the mica spectra, also lie within the range of the calculated spectrum and are labeled in Fig. 12. These calculations clearly illustrate the effect of temperature and density on the line and continuum radiation intensity over a range of T_e and N_i that might occur in the stagnating plasmas in these experiments.

As discussed in Sec. II B, the unknown reflectivity of the crystals and the difficulties associated with measuring this quantity make it impossible to make absolute measurements of the intensity of any spectral feature, despite the fact that the DEF film used to image the spectrum has a known spectral response. Therefore, it is not meaningful to compare the absolute intensities predicted by these calculations with what is observed on the mica spectra. However, the way in which the intensities of the various spectral features predicted by the calculations change relative to one another as temperature and density are varied can prove extremely useful in trying to understand the mica spectra. It is instructive to compare two sets of calculated spectra: In the first set, the electron temperature is maintained at 300 eV while the density is varied. In the other set, the temperature is varied while density is maintained at 10^{20} cm^{-3}. These values of constant temperature and density are reasonable values for the wire arrays used in these experiments (explained below) and are sufficient for the production of Al XIII in both sets of spectra.

Consider the intensity of the Al XIII δ line and the continuum radiation immediately surrounding it. In order to observe emission from this line and also continuum radiation at one axial position on the spectrum, but for both sources to be absent at a different axial position, there must be a variation in either density or temperature (or both) between the two positions that results in a drastic change in intensity in both radiation sources. In Sec. I it was described how the implosion phase of a wire array leaves behind up to 40% of the original mass at the initial wire positions, so that ~60% of the original mass is assembled on axis at the time of x-ray peak. Note that the trailing mass that remains near the initial wire positions has a highly modulated distribution (i.e., the 40% mass that makes up the debris field is not evenly distributed in the axial direction) and so some regions of the debris field may contain almost as much mass as they did prior to implosion, while other regions may contain close to zero mass. For a 1 mm diameter stagnating column, N_{max}, the maximum possible number density per cubic centimeter,

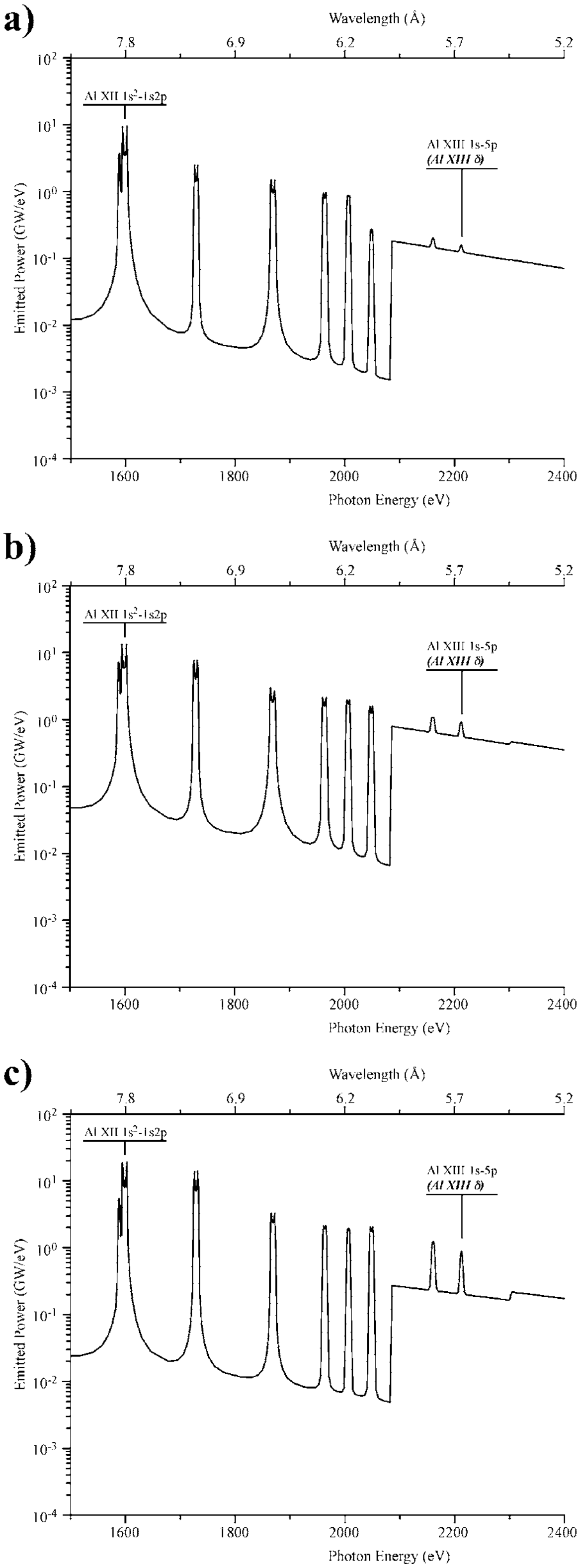

FIG. 12. Calculated spectra with (a) T_e=300 eV and $N_i=10^{20}$ cm^{-3}. This density corresponds to $N_i=0.5N_{\max}$, where $N_{\max}$ is the maximum possible number density for the arrays used in these experiments, (b) T_e=300 eV and $N_i=N_{\max}=2\times10^{20}$ cm^{-3}, (c) T_e=390 eV and $N_i=0.5N_{\max}=10^{20}$ cm^{-3}.

occurs when the column contains 100% of the wire material. $N_{\max}$ has a value of $\sim2\times10^{20}$ cm^{-3} for the arrays used in these experiments.

Consider first maintaining constant temperature (T_e=300 eV) and the intensity of the continuum radiation immediately surrounding the Al XIII δ line. At a density of 50% N_{max} (close to the 60% proportion of the original mass observed to participate in a typical implosion), the continuum intensity is $\sim1.25\times10^{-1}$ GW/eV, shown in Fig. 12(a). Now, consider a region of the stagnated column with an above average density, for example, of 2×10^{20} cm^{-3}, which corresponds to 100% $N_{\max}$ [shown in Fig. 12(b)]. The intensity of the same continuum radiation increases to $\sim6.3\times10^{-1}$ GW/eV; a factor of ~5 increase from the 50% $N_{\max}$ case. For the Al XIII δ line, maintaining constant temperature as the density is varied from 50% $N_{\max}$ to 100% $N_{\max}$ produces, respectively, an increase in intensity from $\sim1.6\times10^{-1}$ to $\sim8.0\times10^{-1}$ GW/eV. Again, a factor of ~5 increase.

Now, if instead of varying density, the temperature is allowed to change while N_i is fixed at 50% $N_{\max}$. This change in temperature will increase the $\sim1.6\times10^{-1}$ GW/eV of Al XIII δ line radiation seen in Fig. 12(a) to $\sim8.0\times10^{-1}$ GW/eV. The conditions that produce this factor of ~5 intensity change through temperature increase are shown in Fig. 12(c), in which T_e=390 eV. When this factor of ~5 increase in the line intensity was produced by varying the density, the intensity of the continuum radiation was also seen to change by a factor of ~5. However, varying the temperature to produce the factor of ~5 increase in the line intensity only produces a factor of ~1.6 increase in the intensity of the continuum radiation. This can be seen between Fig. 12(a) and Fig. 12(c), where the continuum immediately surrounding the Al XIII δ line only increases from $\sim1.25\times10^{-1}$ to $\sim2.0\times10^{-1}$ GW/eV as temperature increases from 300 to 390 eV, less than one-third the change observed when density was increased.

These calculations demonstrate that the production of line radiation is equally as sensitive to density change as it is to temperature change over a range of T_e and N_i that are applicable to these experiments. However, while the production of continuum radiation is equally sensitive to density as the production of line radiation, it shows a relative lack of sensitivity to temperature change. Put another way, for an equal factor of gain in Al XIII line radiation from either temperature or density increase, a density increase will be accompanied by a much greater gain in continuum radiation.

From the calculated spectra shown in Figs. 12(b) and 12(c) it is clear that the only feature of the spectrum that shows an obvious difference between the effects of high density and high temperature is the ratio of the Al XIII lines to the continuum immediately surrounding them: The relative intensities of all the other lines in the calculated spectra do not vary sufficiently between the high-density and high-temperature cases to be useful. However, the contrast between the Al XIII lines and the continuum is much greater for high temperature than it is for high density, and thus an experimental measurement of this contrast would go some way to determining whether temperature or density is dominant in the bright spots. From the results obtained from time-integrated mica crystal spectra, it is difficult to assess to what degree the bright spot emission is a result of high density as opposed to high temperature. This is due to the fact that mica

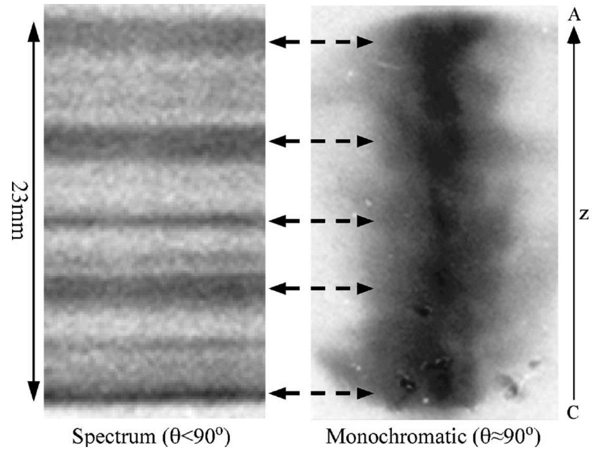

FIG. 13. A comparison of the axial position of continuum radiation in both a crystal-produced spectrum and a monochromatic image (at ≈6.65 Å).

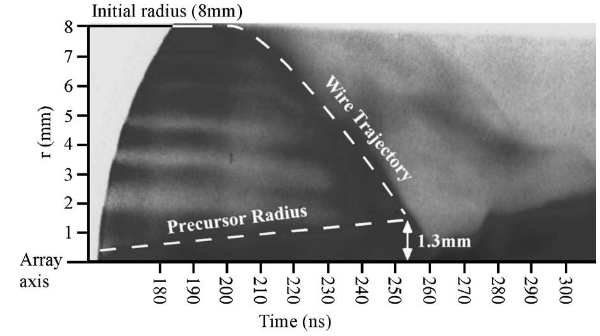

FIG. 14. Streak-camera image showing the expansion of the precursor plasma and the wire trajectory during the snowplough implosion. The imploding sheath is seen to impact upon the precursor at a radius of 1.3 mm.

reflects in multiple orders, hence making it impossible to make an absolute measurement of the contrast between the line radiation and the continuum. The matter is further complicated by the fact that, as shown in Fig. 9, different orders are subject to different levels of transmission through the debris shielding and uv-visible filters. Future experiments will use the single order reflection from quartz to make a measurement of the contrast between the Al XIII lines and the continuum radiation, but existing data and calculations limit the conclusion to a strong possibility that high density is responsible (at least in part) for the bright spot emissions.

Figure 13 shows a time-integrated monochromatic image from the quartz crystal and a section of the $R=186$ mm mica crystal spectrum containing only continuum radiation, both from the same shot. The stagnated column, as shown by the monochromatic image, is made up of several intensely radiating regions which appear close to the axis, and the axial position of these regions clearly matches the position of the continuum regions on the spectrum. Both the mica spectra and the monochromatic images capture the entire length of the pinch, and so aligning axial features between the two is a simple matter. For the purpose of identifying from which radial positions continuum radiation originates, it would be ideal if it were possible to produce a monochromatic image purely in continuum radiation. Unfortunately, monochromatic imaging of aluminum arrays with quartz crystal produces an image at ≈6.65 Å, which lies close to the He-like $1s3p$ line (He β). Despite the very narrow wavelength range captured when imaging at a Bragg angle so close to 90°, it is very possible that the wing of this nearby line might be captured. However, the following two observations must be noted: First, the axial position of the continuum radiation imaged by the mica crystal matches extremely closely the axial positions of the intense features observed in the monochromatic images. Secondly, the same intense features are observed at the same locations on all the images obtained with the time-integrated pinhole cameras, despite the various filters used, indicating that these features must be emitting continuum radiation. Therefore, despite the possible presence of line radiation, it is most likely that the intense features observed on the monochromatic images are, dominantly, a result of continuum radiation from bright spots. It is possible that the nearby He-like line might be responsible for the "halo" of radiation that can be seen surrounding the bright spots on the monochromatic image in Fig. 13, particularly as this feature on the image is present along the entire length of the pinch.

The radius of this bright spot region is ≈1.4 mm at the widest point. Figure 14 is an optical streak-camera image (from the same shot as Fig. 13) which shows how the radius of the precursor plasma and the position of the imploding sheath (the wire trajectory) change over time. From this image it can be said that the imploding sheath arrives at the outside of the precursor column at a radius of ≈1.3 mm. The similarity between this radius and the radial extent of the bright spot region in Fig. 13 suggests that the size of the region in which bright spots are formed might be influenced by the radius of the precursor at the time the imploding sheath impacts upon it.

The XUV framing cameras used in these experiments are sensitive to radiation above ≈30 eV when unfiltered and thus, being sensitive to this low-energy radiation, produce images that provide information on the bulk, relatively cold, plasma dynamics. An example of an image produced by these cameras is shown in Fig. 15(c), taken 14 ns prior to x-ray peak, in which a dense column of plasma can be seen

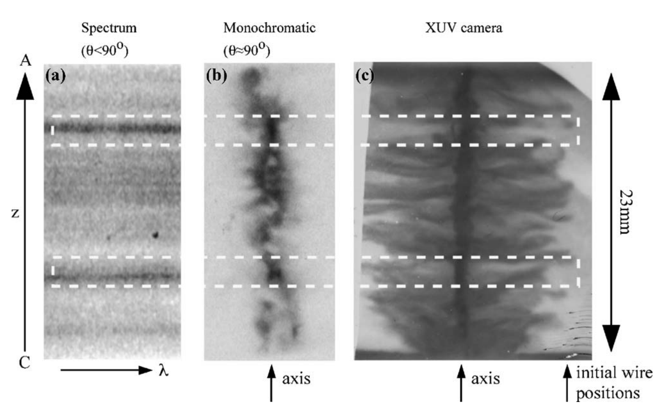

FIG. 15. The correlation between the axial position of bright spots imaged by (a) mica crystal operating in the $\theta<90°$ regime as well as (b) quartz crystal operating in the monochromatic regime ($\lambda\approx6.65$ Å) and the axial position of "swept out" regions imaged by (c) time-resolved XUV pinhole photography at 266 ns (14 ns prior to x-ray peak).

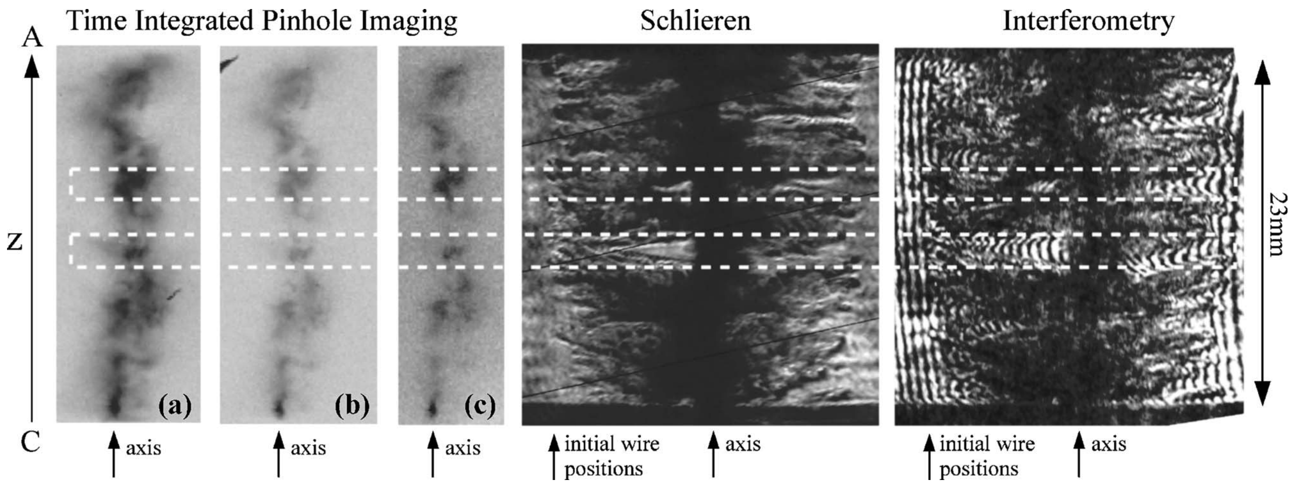

FIG. 16. The correlation between the axial position of bright spots in time-integrated pinhole images filtered with (a) 20 μm Be, (b) 12.5 μm PVDC +1 μm Al, (c) 15 μm Si, and the axial position of "swept out" regions imaged by time-resolved Schlieren and interferometry at 275 ns (3 ns prior to x-ray peak).

on axis while filaments of trailing mass stretch from the axis out to the initial wire positions. This XUV image is compared to two other images captured in the same shot: Fig. 15(a), a section of the continuum radiation from a time-integrated spectrum produced by the $R=186$ mm mica crystal; and Fig. 15(b), a time-integrated monochromatic image in, dominantly, continuum radiation. Comparing these images clearly reveals a correlation between the axial position of bright spots in images (a) and (b) and points along the axis at which very little debris remains, shown by the XUV image. This lack of debris could well result in a higher plasma density in the pinch column at these points, as more mass is concentrated on axis at certain axial positions.

Figure 16 shows another similar comparison, this time between time-integrated pinhole images (which show the position of bright spots in much the same manner as monochromatic images) and time-resolved images from laser probing; an interferogram and Schlieren image both captured ≈ 3 ns before the peak of the x-ray pulse. Although the filters used on these pinhole cameras permit line radiation to pass, the intensely radiating regions have been shown to be continuum-emitting bright spots by comparison with crystal imaging (shown for this same shot in Fig. 13). Figure 16 shows that, again, the bright spots on these time-integrated pinhole images correlate with positions along the axis at which nearly all the plasma has reached the stagnating column, while at other locations that show an absence of bright spots there still remains a great deal of material yet to arrive at the axis. It should be noted that occasionally a bright spot is seen at an axial position where there remains a significant amount of trailing mass. However, it is possible that these bright spots do correspond to a lack of trailing mass, but that this lack of debris occurs along an implosion path close to the line of sight on which the 2D images are captured, and is therefore not possible to see. In any case, there are very few bright spots that are difficult to correlate to a lack of trailing mass, and where an obvious lack of debris is observed, such as in Figs. 15 and 16, bright spots are always seen at these axial positions.

The position of the axis on time-integrated pinhole images (such as those in Fig. 16) is deduced from comparison of the pinch shape with other diagnostics. For example, the stagnating column in Fig. 16 shows a clear kink towards the top of the image, and this can be matched to the same shape in the interferogram. Deducing the position of the axis from interferometry is a simple matter, as an interferogram of the array prior to firing is obtained for every shot in order to provide an image of the fringe pattern with zero plasma density.

In experiments it has been observed that, at the start of the implosion, the quasiperiodic wire breakage seeds an instability that results in the leading edge of the imploding sheath becoming modulated.[28,29] This global instability grows in both amplitude and wavelength; the amplitude increases until it spans the entire radius of the array, producing tendrils of plasma reaching from the initial wire positions all the way to the axis, as shown in Fig. 15. The appearance of this global instability might be described as an "$m=0$-like" instability growing from a "Rayleigh-Taylor-like" instability. From Figs. 15 and 16 it is clear that by the time the imploding sheath arrives at the precursor, the global instability has produced a severe modulation in the arrival of mass on axis: At times close to the peak of the x-ray pulse, the majority of the wire material has arrived at the precursor at some axial positions, while at others there remains a significant amount of material between the axis and the original positions of the wires.

It is possible to imagine an explanation for the correlation between bright spots and swept out axial positions that involves a localized delivery of kinetic energy with the arrival of an above average (over the length of the pinch) amount of mass on axis. This might produce either a localized compression of the pinch, resulting in increased density at certain points, or thermalization to high temperature in these regions. However, if bright spots are a result of the kinetic energy delivered on axis as swept out regions develop, then the continuum radiation would be expected to commence early, at the very start of stagnation, prior to the peak of the x-ray pulse. Also, since the kinetic energy explanation effectively involves an impact on the precursor, the continuum radiation might also be expected to be short-lived relative to the main x-ray pulse. As will be discussed later in greater detail, time-resolved measurements of the continuum radiation reveal that although it is short-lived relative to the main pulse, it does not start early, prior to the peak of the main x-ray pulse.

An alternative explanation for the correlation between bright spots and "swept out" regions is that, at positions where little debris remains, the only path for the current to take is close to the axis, resulting in greater $J\times B$ compression (and possibly heating) than at locations where the current may be dispersed throughout debris, away from the axis. In this model, the appearance of bright spots is not due to a concentration of kinetic energy at particular axial positions, but a concentration of current and thus compression to higher density. There are several additional mechanisms that might arise as current is concentrated into a small region near the axis: There is the possibility of increased ohmic heating, and also of MHD instabilities (such as the $m=0$ sausage instability, the $m=1$ kink instability, and higher orders), which may result in localized pinching ($m=0$) and twisting ($m=1$). The onset of $m=0$ instabilities is particularly interesting since this might result in localized compression to high densities and therefore increased continuum radiation production. Evidence for the presence of $m=0$ instabilities in wire arrays and a correlation between their position and that of bright spots is discussed later.

B. Radial structure from 2D imaging

Monochromatic images in continuum radiation, such as that shown in Fig. 13, typically show a number of strongly radiating bright spots distributed close to the axis in the radial direction. From these images it was possible to measure the maximum distance from the axis that bright spots were located for each shot. This distance, ΔR_{max}, then defines the radius of a cylinder—centered on the array axis—within which all bright spots are located. For these arrays, with 32 wires of 10 μm aluminum, values of ΔR_{max} ranged from 1.4 to 1.9 mm, with a mean value of 1.6 mm. Again, this range of radii is very similar to the radius at which the imploding sheath impacts upon the precursor. Bott *et al.*[30] have shown that for arrays of this mass and radius at the same current level, at the time the imploding sheath arrives, the radius of the precursor plasma can be expected to be $\approx$1.25–1.5 mm. Optical streak-camera photography, such as shown in Fig. 14, agrees well with these sizes.

Bland *et al.*[31] have shown that the imploding sheath is actually composed of a leading and trailing edge, and so the time of impact of the imploding sheath upon the precursor discussed here refers to the time at which the leading edge impacts. The results in Ref. 31 go on to show that the main x-ray pulse starts to take off as this leading edge hits the precursor, and reaches peak at, or just after, the arrival of the trailing edge, perhaps explaining the rise time of the x-ray pulse as the temporal width of the imploding sheath. For the arrays discussed in this paper, the peak of the x-ray pulse does not occur until typically ~15–20 ns after the impact of the leading edge of the imploding sheath upon the precursor,

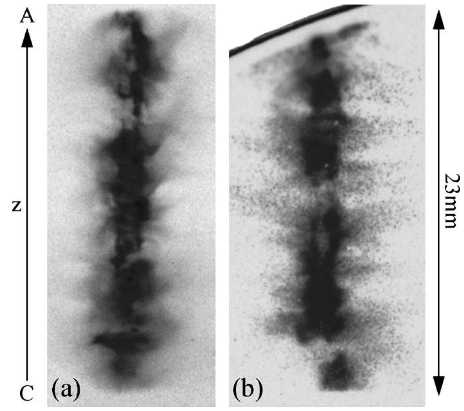

FIG. 17. (a) Time-integrated pinhole image filtered with 6 μm Al showing reduced intensity on axis, suggesting a "hollow" radiation structure and (b) similar radiation structure from a time-resolved pinhole image at 269 ns (x-ray peak at 271 ns) filtered with 7 μm Be.

by which time the pinch has been compressed to a smaller radius. Measurements of the pinch from XUV camera images captured close to x-ray peak show a maximum compression to a radius of less than 0.5 mm. The volume of the pinch is considerably smaller at the time of x-ray peak than at the time when the leading edge of the imploding sheath impacted upon the precursor. From the monochromatic imaging results described above, a significant fraction of the intensely emitting bright spots are located outside this volume. However, as mentioned previously, the similarity in size between the volume occupied by the precursor at the time of impact and the volume occupied by the bright spots does raise the question of whether the precursor diameter at the time of impact affects where bright spots are formed.

Time-integrated pinhole camera images for these same arrays suggest a possible "hollow" radiating structure to the stagnating column: The center of the pinch appears to radiate less strongly than the edges. An example of this is shown in Fig. 17(a), which shows the most intense radiation originating from regions located further than $\approx$0.5 mm from the axis. The same type of structure is visible in time-resolved pinhole images filtered with 7 μm beryllium (passes radiation $>$500 eV), an example of which is shown in Fig. 17(b). This image, captured 2 ns before the peak of the main x-ray pulse, clearly illustrates the reduced intensity on axis at some axial positions, although the 7 μm beryllium filter lets both line radiation and continuum pass and so it is not possible to conclude which areas are intense continuum emitters (bright spots) and which are simply emitting strong line radiation.

Multiple-angle time-integrated pinhole imaging from two cameras allows a comparison of the position and shape of bright spots and strongly radiating regions from two different positions, 90° apart azimuthally. Figure 18 shows examples of these pairs of images from two different shots, both 32 wire arrays of 10 μm aluminum. In each pair, some bright spots are present in both images but appear to shift horizontally or change shape, which indicates that these features do not lie along the axis but rather some distance away from it so that they appear to be in different places when viewed from a different angle. Also, some bright spots are present on one image, but do not appear on the other image, which could be interpreted as the feature being located to the side of the central column from one viewpoint but behind it from another. This indicates both that the bright spot is located away from the axis, and also that the plasma on axis is, to some degree, opaque to bright spot radiation and is therefore able to "hide" a particular feature if it lies behind the plasma column with respect to the camera. These observations lend further weight to the argument that the stagnated column has a hollow radiation structure, with reduced intensity on axis. It is important to note the difference between a hollow radiation structure and a hollow density profile: Although the stagnated column might be observed to emit more strongly close to the surface, this does not immediately imply anything about the density near the edges compared to the interior.

It has already been suggested from Figs. 15 and 16 that the positions of these bright spots correspond to axial sections of the array that have imploded to leave relatively little debris, and that this might result in a concentration of current at the axis. It would now appear that if this is the case, the current does not flow on axis, nor cause bright spots to develop in an azimuthally symmetrical manner. It might be argued that the current path in these swept out regions lies towards the outside of the stagnated column, and that the mechanism by which it produces bright spots (be it through compression or heating) does not act uniformly around the circumference of the column, but appears concentrated at certain azimuthal positions. Previous observations of bright spots in wire arrays have been attributed to instabilities or electron beams (e.g., experiments by Deeney *et al.*[32] on the Double-EAGLE generator with 12-wire aluminum arrays), not least because it is the $m=0$ instability mechanism that produces bright spots in single wire Z pinches and X pinches.

In the case of single wires and X pinches, the mechanism by which instabilities give rise to bright spots involves an azimuthally symmetric, pure $m=0$ mode. It would now appear that for bright spots in wire arrays this is not the case. If MHD instabilities are responsible for the development of bright spots in wire arrays, it is not the classic azimuthally symmetric pure $m=0$ mode. It may be that current concentrated around the edges of the stagnating column generates MHD instabilities along the circumference of this column. Images such as Figs. 17(a) and 18 suggest that several bright spots can occur at one axial position but be separated azimuthally. This implies that at some points on the axis there might be either several instabilities on the circumference at one time, or (since these images are time-integrated) that several instabilities occur at different points on the circumference at one axial position but at different times during stagnation.

Experiments with copper wire arrays on the MAGPIE generator have provided evidence that the axial positions at which bright spots occur involve the presence of electron beams. The 186 mm mica crystal was used to produce a spectrum of an $N=8$ array of 10 μm copper, which is shown in Fig. 19. At a Bragg angle of 38° the K_α doublet ($\sim$1.54 Å) is imaged as an eighth-order reflection, while

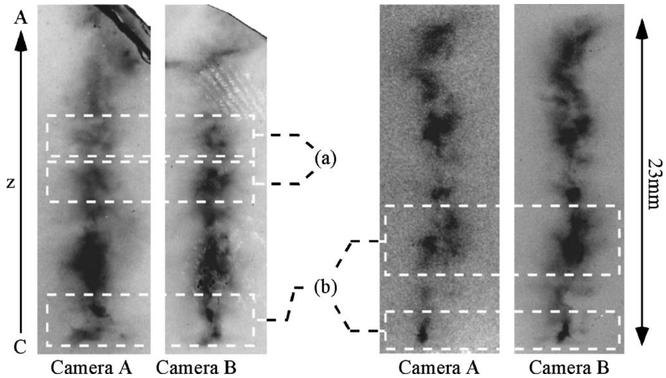

FIG. 18. Pairs of images from two time-integrated pinhole cameras (20 μm pinholes in camera A, 50 μm pinholes in camera B) both filtered with 15 μm silicon. Some features (a) are present in one image, but not in the other, while other features (b) are present in both images but appear shifted due to the rotation of perspective.

other softer spectral lines from Ne-like copper (Cu XX) appear as a result of first-order reflections. That the K_α line is seen is a clear indicator of the presence of an electron beam, since this radiation is a result of a collision between a high-energy electron and an ion: The collision results in the ejection of a *K*-shell electron from the ion, followed by the emission of a photon as an *L*-electron makes a transition into the empty state. The mechanism by which K_α photons are produced means that this radiation originates from relatively "cold" regions of plasma that is not highly ionized, whereas the other spectral lines that are present originate from highly ionized species and therefore from much hotter and/or denser parts of the plasma. From Fig. 19 it is immediately clear that none of the lines in this spectrum are present along the entire length of the array: The Cu XX lines are only present at axial locations where continuum is also present, the same effect as in the aluminum spectra (i.e., bright spots). K_α radiation is also seen faintly in these continuum regions, but is much more intense at axial locations where there is no continuum radiation: While the Cu XX lines are only present at three locations along the pinch (at the electrodes and at one point near the center), the K_α radiation is most intense in the two regions that lie between these bright spots. One possible explanation for this anticorrelation is that some mechanism in the bright spot regions (perhaps the formation of $m=0$ instabilities) produces an electron beam that then impacts upon the cold plasma adjacent to it, producing the K_α radiation. Another possibility is that electron beams are formed in the

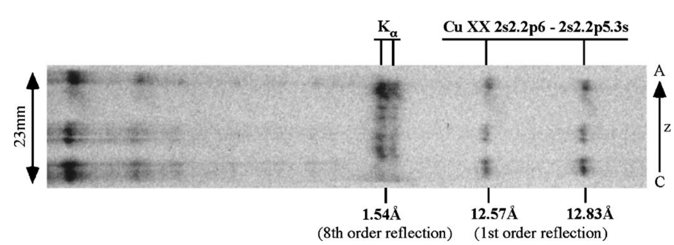

FIG. 19. The spectrum from a copper array, showing the K_α doublet in eighth-order reflection and a number of other spectral lines as a result of first-order reflections.

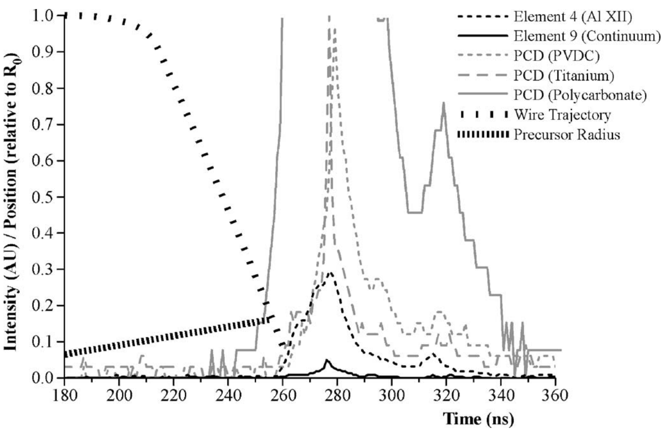

FIG. 20. X-ray intensity measured by PCDs (titanium filter to block Al XII *K*-shell lines, PVDC filter to pass Al XII *K*-shell lines) and detector array elements (element 4 predominantly observing an Al XII *K*-shell line, element 9 predominantly observing continuum radiation).

regions where the K_α radiation is seen, and then impact upon a dense region of plasma, causing heating and compression that results in the bright spots. The former explanation proposes that the bright spots produce the electron beams, while the latter suggests that the electron beams produce the bright spots. Future experiments on the MAGPIE generator will seek to address the issue of which mechanism holds the most validity, or if a different explanation altogether is required.

C. Time-resolved spectra

Imaging of the spectrum produced by the $R=100$ mm mica crystal onto the 10-element silicon detector array allowed time-resolved intensity measurements of the spectral windows defined by the width and position of each element. The array captures $\approx 1/3$ of the length of the pinch, and so the axial spread of the continuum regions often results in only one very intense continuum region being captured by the array. This means, in the case of the spectrum shown in Fig. 7, that it is possible to discuss the signal from the array elements in terms of one particular bright spot. Had the entire spectrum been focused down so that the detector array captured the entire length of the pinch, it would only have been possible to discuss an average over all the continuum regions.

Comparison of the two lineouts in Fig. 11 shows that at wavelengths where Al XII *K*-shell lines are present, the intensity of the line radiation dominates over the continuum. Similarly, in areas of the spectrum where there is continuum radiation and Al XIII line radiation only, the lineouts show that the intensity of the continuum dominates over the line radiation. Therefore, from the discussion in Sec. III A regarding the presence of Al XIII lines in continuum regions and their absence in noncontinuum regions, elements of the detector array that are positioned to intercept radiation from Al XII lines are observing predominantly the colder, bulk plasma consisting of species less ionized than Al XIII. Elements that intercept only continuum and Al XIII line radiation are observing bright spots.

Figure 20 shows the x-ray intensity measured by three PCDs and two elements of the silicon array. The PCDs were filtered, one with 12.5 μm titanium (passes 2–5 keV radiation) which blocks the Al XII *K*-shell lines but passes continuum, another with 12.5 μm PVDC (passes 1.1–2.8 and >2.8 keV radiation) which allows the Al XII *K*-shell lines to pass, and the third with 6 μm polycarbonate (passes 200–300 eV and >800 eV radiation) which allows both lines and continuum to pass as well as soft radiation. The two array elements were aligned as shown in Fig. 7, so that element 4 observed the radiation from the Al XII resonance line, while element 9 predominantly observed continuum radiation.

Within experimental errors of ± 2 ns, all five signals shown in Fig. 20 were seen to peak at the same time (at ≈ 275 ns). However, both the *K*-shell radiation and the continuum radiation begin to rise at a significantly later time than the soft radiation, measured by the 6 μm polycarbonate PCD; confirmation that both *K*-shell and continuum radiation are not produced in significant amounts during the snowplough phase of the implosion, but later, when the imploding sheath reaches the precursor column.

Despite the peaks of both continuum and line radiation coinciding, the duration of the main pulse is decidedly different for the two types of radiation: The full width at half maximum (FWHM) for the line radiation pulse and the continuum pulse as measured by the array elements is ≈ 19 ns (element 4) and ≈ 5 ns (element 9), respectively, with a time resolution of ≈ 1 ns. The PCDs measured the line and continuum FWHM to be ≈ 9 ns (PVDC filter) and ≈ 3 ns (titanium filter), respectively, with a time resolution of ≈ 1 ns. The difference in value of the FWHM for line radiation measured by element 4 and the PVDC filtered PCD is probably due to the different range of wavelengths observed by these two different instruments, and the same can be said for the two values of FWHM for continuum radiation. The important feature of these measurements is that when the line and continuum FWHM was measured by the same instrument (silicon array elements or PCDs), in both cases the FWHM of the line radiation was observed to be three to four times longer than for the continuum radiation. This indicates that the bright spots achieve the conditions that result in the production of strong continuum and Al XIII line radiation for a significantly shorter time than the bulk plasma can sustain Al XII line radiation. Hence, the mechanism responsible for the creation of bright spots is in operation and peaks at the same time as the main x-ray pulse, but is only active for a much shorter period of time than the mechanism(s) that heat the bulk plasma up to temperatures needed for the production of Al XII line radiation.

As discussed in Sec. III A, the fact that the continuum radiation is short-lived is consistent with the idea that bright spots are created by a rapid delivery of kinetic energy that may be thermalized or result in compression. However, as can now be seen from Fig. 20, the timing of the continuum radiation is not consistent with an explanation based on the thermalization of kinetic energy. The wire trajectory, measured from radial streak photography (see Fig. 14), shows that the latest arrival of the imploding sheath at the outside of the precursor occurs at ≈ 255 ns (the radial streak camera samples a narrow axial section of the wire array, making it likely that the wire trajectory and timings obtained from Fig. 14 pertain to the bulk implosion rather than the swept out regions), while the continuum (and other radiation) peaks at $\approx$275 ns. It is unlikely that the thermalization of a large amount of kinetic energy upon impact with the precursor would produce a radiation pulse that only peaks 20 ns later. Under conditions that might be expected to exist in the swept out regions, calculations show that the thermal equilibration time is, at the very longest, always less than a nanosecond. At an electron temperature of 340 eV and electron density of 10^{21} cm^{-3}, the ionization time from the Al XI ground state to that of Al XII is less than 0.1 ns. Typical collisional excitation times to the $n=2$ levels of Al XII from the ground state are 2 ns. Ionization from Al XII to Al XIII is dominated by ionization from excited states (the ladder process discussed previously), and is in total about equal to the 2 ns excitation time, since the ionization time from the excited states is well under 1 ns. In total, this is much less (by approximately an order of magnitude) than the 20 ns delay observed between the arrival of the implosion sheath and the peak of continuum (and other) radiation.

If the bright spots were a result of current becoming concentrated on axis, discussed in Secs. III A and III B, then this delay might be interpreted as the growth time (once current has arrived) for $m=0$ instabilities to form around the precursor and produce the conditions required for bright spot radiation.

IV. CONCLUSIONS

Spherically bent mica crystals have been used to produce time-integrated spectra of imploding aluminum wire arrays with axial resolution. These spectra reveal that the stagnated pinch exhibits "bright spots," i.e., narrow axial regions that emit intense continuum radiation. Strong emission from Al XII lines is observed along the entire length of the pinch, whereas radiation from Al XIII is only seen in the bright spots, where continuum radiation is also present. Calculated spectra from detailed atomic models illustrate that both increased temperature and density could produce Al XIII line radiation, but that the presence of intense continuum radiation makes high density a more likely explanation for the bright spots.

Comparison of crystal imaging with time-resolved XUV imaging, and laser probing with time-integrated pinhole imaging, reveals a correlation between the axial positions of bright spots and where the global instability has modulated the imploding sheath to leave very little trailing mass. A possible explanation is that in these regions where little debris remains, the only available current path is close to the axis, resulting in greater compression and heating than at positions where the current path could be dispersed throughout the low-density plasma away from the axis. At these debris-free positions, a concentration of current might possibly lead to the formation of MHD instabilities.

Monochromatic crystal imaging has produced images showing bright spots located in a region of similar size to that occupied by the precursor plasma at the time the imploding sheath impacts, suggesting that the size of the precursor at the time of impact may influence where bright spots are formed.

Both time-integrated and time-resolved pinhole imaging reveal a "hollow" radiation structure of the stagnating column, with reduced intensity on axis. Time-integrated pinhole cameras separated azimuthally by 90° have produced pairs of images in which bright spots change horizontal position relative to the axis and are obscured or change shape between images. This indicates that these features are located away from the axis and also that the plasma on axis may be opaque to bright spot radiation. This, combined with the presence of bright spots at "swept out" axial regions, indicates that if some fraction of the current is concentrated near the axis, it lies towards the outside of the stagnated column, and that the distribution of current might not be azimuthally uniform. These pairs of azimuthally separated images suggest the possibility of localized MHD instabilities occurring at different points around the circumference (but at the same axial position) of the stagnating column.

Imaging of the axially resolved spectrum produced by a spherically bent mica crystal onto the elements of a linear silicon x-ray photodiode array provided time-resolved measurements of the intensity of line emission and continuum emission. Both line and continuum radiation intensity were observed to peak simultaneously, however the FWHM of the line radiation was measured to be $\sim$3–4 times longer than that of the continuum. From this, it is possible to conclude that the mechanism responsible for producing continuum and Al XIII line radiation operates efficiently for a much shorter time than that required to produce Al XII line radiation. Finally, the timing of peak continuum radiation power (observed by the photodiode array) compared to the timing of the implosion trajectory suggests that the bright spots present in the "swept out" regions are a result of current concentration near the axis as opposed to a localized thermalization of kinetic energy. The thermalization of kinetic energy into radiation is likely to occur in less than 1 ns, whereas a delay of 20 ns was observed between the impact of the imploding sheath on the outside of the precursor column and the peak of continuum radiation power. This 20 ns delay might be interpreted as the time taken for the mechanism responsible for bright spot production to peak once the "swept out" regions develop, such as the growth time of MHD instabilities brought on by the concentration of current near the axis.

Future work on the MAGPIE generator will involve a combination of time-resolved imaging of the entire spatially resolved spectrum produced by spherically bent crystals with measurements of array dynamics to further understand the physical processes involved in x-ray production. Development of imaging techniques using spherically bent crystals will address the issue of whether increased density or temperature dominates the production of bright spot radiation.

Different wire materials produce precursor plasmas of different radii from aluminum, and since the radius of the precursor column appears to play a role in the size and position of the emitting regions at stagnation, further investigation into how the structure at stagnation varies with wire material is required. There is the possibility of using extremely thin wires to move the stagnation phase very early in time, before the precursor column has time to form, which would produce an imploding sheath that impacted upon itself at the axis, not a precursor. A drastic change in emission characteristics and stagnation structure under this regime would show that a study of the effects of precursor characteristics upon x-ray production at stagnation is required.

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